



Quantifying responses of dung beetle assemblages to cattle grazing removal over a short-term in introduced Brazilian pastures

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ABSTRACT

Pasture management techniques may affect the biodiversity of insects beneficial to pastures, such as dung beetles. Cattle grazing removal over a short-term is widely used in introduced Brazilian pastures. However, the impact of this management on dung beetles is still unknown. This study evaluated the taxonomic (species richness, abundance, biomass, species composition, Shannon, and Simpson indexes) and functional (functional richness FRic, functional dispersion FDis, functional evenness FEve, and community-weighted means of trait values CWMs) dung beetle assemblages response to cattle grazing removal over a short-term in introduced pastures. We sampled dung beetles, with pitfall traps baited with cattle dung and carrion, simultaneously in pastures with constant cattle grazing (control), and one, three, and five months after cattle grazing removal in Aquidauana, Mato Grosso do Sul, Brazil. Taxonomic metrics and FRic, FDis, and FEve did not differ among control and pasture treatments. We found that pastures with one month of cattle removal maintain the same CWM coprophagous diet from the control, and higher CWM biomass in relation to control. Although cattle grazing removal, at least over the short-term, does not cause a negative impact on dung beetle taxonomic metrics and functional diversity indexes, two important traits associated to dung removal are negatively affected after three months of cattle removal. Thus, we suggest that one month of cattle removal to grazing rotational management in introduced Brazilian pastures can be a useful strategy to, besides conserving the grass, also conserving dung beetle diversity, and consequently the ecological functions performed by them.

1. Introduction

In the Neotropical region there are many pastures planted with exotic grass species that are used for raising herbivorous mammals such as horses, goats, sheep and especially cattle. Brazil has an estimated area of 172 million hectares of native and cultivated pastures, with roughly 99 million hectares planted pastures (IBGE, 2017). The exotic grass *Urochloa* spp. (Poaceae) occupies about 80% of the entire cultivated pasture area in the country (Galdino et al., 2016). This African grass was introduced into Brazil over two centuries, presenting excellent adaptation to the local climate and soil characteristics, becoming widely used for cattle production (Pereira et al., 2005).

Pastures of *Urochloa* spp. grass, typically after 4–10 years of grazing, undergo a sharp decline in productivity, as a result of inadequate use of high stocking rates and lack of maintenance fertilization (Spavorek et al., 2007). Overgrazing and soil nutrient deficiency are the main causes of degradation of these pastures (Spavorek et al., 2007; Galdino et al., 2016). Thus, management of these grasses is important to maximize forage production and stabilize it for animal consumption (Boval and Dixon, 2012). One of the most commonly used practices in cattle production is rotational grazing management, which consists of different periods of grazing and abandonment. Continuous grazing management is also a widely used practice; however, the effects of integral occupation may harm regrowth of the forage (Eaton et al., 2011).

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Indeed, correct use of pasture management practices not only increases forage productivity and animal consumption (Boval and Dixon, 2012), but may also reduce greenhouse gas emission (Mazzetto et al., 2015) and exert a form of control on the development of arthropod pests, for both plants and animals (O'Neil et al., 2003; Branson et al., 2006). Likewise, the management techniques employed may also affect the diversity and abundance of insects beneficial to pastures, such as dung beetles (Coleoptera: Scarabaeinae) (Correa et al., 2019; Pecenka and Lundgren, 2019). Most dung beetle species bury and utilize the dung for nesting and feeding of offspring (Hanski and Cambefort, 1991), so they increase the nutrient recycling rate in the soil (Menéndez et al., 2016), improve soil fertility and aeration (Bang et al., 2005; Brown et al., 2010), contribute to the reduction of flies (Braga et al., 2012) and parasitic worms (Nichols and Gómez, 2014; Sands and Wall, 2017). In addition, they assist in pasture growth and development (Johnson et al., 2016), increase the grazing area by incorporating the fecal masses (Miranda et al., 2000; Mariategui et al., 2001) and reduce greenhouse gas emission (Penttilä et al., 2013; Slade et al., 2016). Thus, ecological functions performed by dung beetles can reduce the use of financial resources to treat livestock health, and increase grass productivity through nutrient incorporation (Lousey and Vaughan, 2006).

Some authors consider the system “plant – herbivore – coprophagous insects” a complex system, since management must meet the requirements of the plants (e.g., fertilization, regrowth), and consumption of fresh mass by the animals (e.g., grazing time, capacity) without adversely affecting the community of dung beetles on site (Lobo et al., 2006; Verdú et al., 2007). In Europe, for example, moderate grazing has been considered a positive factor for conservation of dung beetle communities (Lobo et al., 2006; Numa et al., 2009; Buse et al., 2015; Tonelli et al., 2018, 2019). In addition, the abandonment of cattle grazing with the suspension of grazing for an extended period and/or irregular spatial distribution for grazing favor the decline in dung beetle diversity (Lobo, 2001; Carpaneto et al., 2007). In the Neotropical region, there are very few studies that have explored the effect of cattle grazing removal on dung beetle communities in introduced pastures. Alvarado et al. (2019) evaluated how livestock intensification affects dung beetle assemblage structure in cattle ranches. Correa et al. (2019) have evaluated the effect of cattle grazing removal on dung beetle communities in natural grasslands (>20 years of abandonment). This type of studies could help cattle farmers in decision-making for a rotational pasture management aiming at, besides forage production, dung beetle biodiversity conservation.

In this study, we evaluated, for the first time, how dung beetles respond (in terms of taxonomic and functional diversity) to cattle grazing removal over a short-term in introduced Brazilian pastures. For this, we sampled dung beetles in introduced pastures (*Urochloa* spp.) with constant cattle grazing (control) and different cattle grazing removal times (one, three and five months) using pitfall traps baited with cattle dung and carrion to answer the following questions: 1) Do species richness, number of individuals, biomass and diversity (Shannon and Simpson indexes) decrease with cattle grazing removal time? 2) Is dung beetle species composition in cattle grazing removal sites different from those in the control? 3) Do functional diversity decrease with cattle grazing removal time? Additionally, we also provide insights into the implications of cattle grazing management on dung beetle conservation in introduced Brazilian pastures.

2. Material and methods

2.1. Area of study

Sampling was conducted in the transition area between the Cerrado and Pantanal ecosystems in Aquidauana, Mato Grosso do Sul, Brazil (20°28'15" S; 55°47'13" W, 150 m a.s.l.). The climate is classified as tropical hot-wet (Aw, Köppen) (Alvares et al., 2014) with a mean annual temperature of 26 °C, an annual rainfall ranging from 1200 to 1300 mm, and a unimodal pluviometric regime with a dry season that goes from

the middle of April to the middle of October and the rest of the year corresponds to the rainy season (Cristaldo et al., 2017). The study site possesses extensive areas of introduced pastures that are used for cattle raising. Overall, pastures are managed in a rotational system where the cattle (1.6 animal unit ha⁻¹) remain for ~ 45 days before being moved to other pastures and the return to the initial pasture can vary after c. a. one, three or five months. All pastures in our study are used for beef cattle production, with often use of veterinary drugs (e.g., ivermectin, usually 0.2 mg/kg an annual dose) and ranging in size from 4 to 50 ha. Regions surrounding the sampling sites are dominated by extensive introduced pastures and patches of native vegetation (Brazilian Cerrado).

2.2. Sampling design

To evaluate the effect of cattle grazing removal, we compared pastures with different cattle grazing removal times: control (constant presence of cattle, $n = 4$), one ($n = 4$), three ($n = 4$) and five months without cattle grazing ($n = 4$). The pastures selected were separated by a minimum distance of 300 m. The distance among pasture sites (300 m) was used to ensure independence of the samples (see da Silva and Hernández, 2015). Therefore, each pasture site were considered a replicate ($n = 4$ sites) for each pasture treatment. In each pasture site we installed one linear transect (100 m) located approximately 50 m from the pasture edge for the allocation of the traps. In the transect we demarcated three points separated by 50 m (Fig. 1). At each point we installed a set of two traps separated from each other by 3 m (see Louzada and Carvalho e Silva, 2009). The traps of each set were baited separately with about 40 g of carrion (rotting bovine stomach) and fresh cattle dung. We used different baits in order to ensure an accurate representation of the local dung beetle functional and trophic groups (see Correa et al., 2016a) and in this way we could also evaluate the trophic preferences of each species (dung versus carrion).

2.3. Dung beetle trapping and taxonomic identification

All sampling occurred simultaneously and was conducted during the rainy season in the study region, in November 2019. The rainy season is the most appropriate period to sample dung beetles in Brazilian pastures (Correa et al., 2018). Dung beetles were sampled with pitfall traps (15 cm diameter and 9 cm deep), installed at ground level, which were partly filled with a solution of 250 mL of water, salt and detergent. Each trap was protected from rain with a plastic lid suspended 20 cm above the surface. The traps remained active for 48 h in each pasture. The baits were placed in plastic containers (50 mL) at the center of each trap using a wire as bait holder. Overall, we installed 12 sets of traps per pasture treatment, with a total sampling effort of 96 traps during the study.

Dung beetles were separated and identified at genus level (Vaz-de-Mello et al., 2011) and then sent to the Universidade Federal de Mato Grosso – UFMT (Cuiabá, Mato Grosso, Brazil) where they were identified at species level by Fernando Z. Vaz-de-Mello. Voucher specimens are deposited in the Entomology Section of the Zoological Collection (CEMT) at the UFMT. To record the biomass of species, all individuals collected were dried (40 ± 5 °C) until we got a constant weight and weighed on a 0.0001 g precision scale.

2.4. Dung beetle traits

We analyzed three functional traits that are directly related to the ecosystem functions performed by dung beetles (Slade et al., 2007; Barrágan et al., 2011): biomass, food relocation behavior, and diet (Table S1). We recorded the biomass of each species. Food relocation behavior was assigned for each species according to the literature (Halfpeter and Edmonds, 1982; Hanski and Cambefort, 1991). Species were categorized as rollers (telecoprids), tunnellers (paracoprids), or dwellers (endocoprids). Roller species remove portions of dung, which

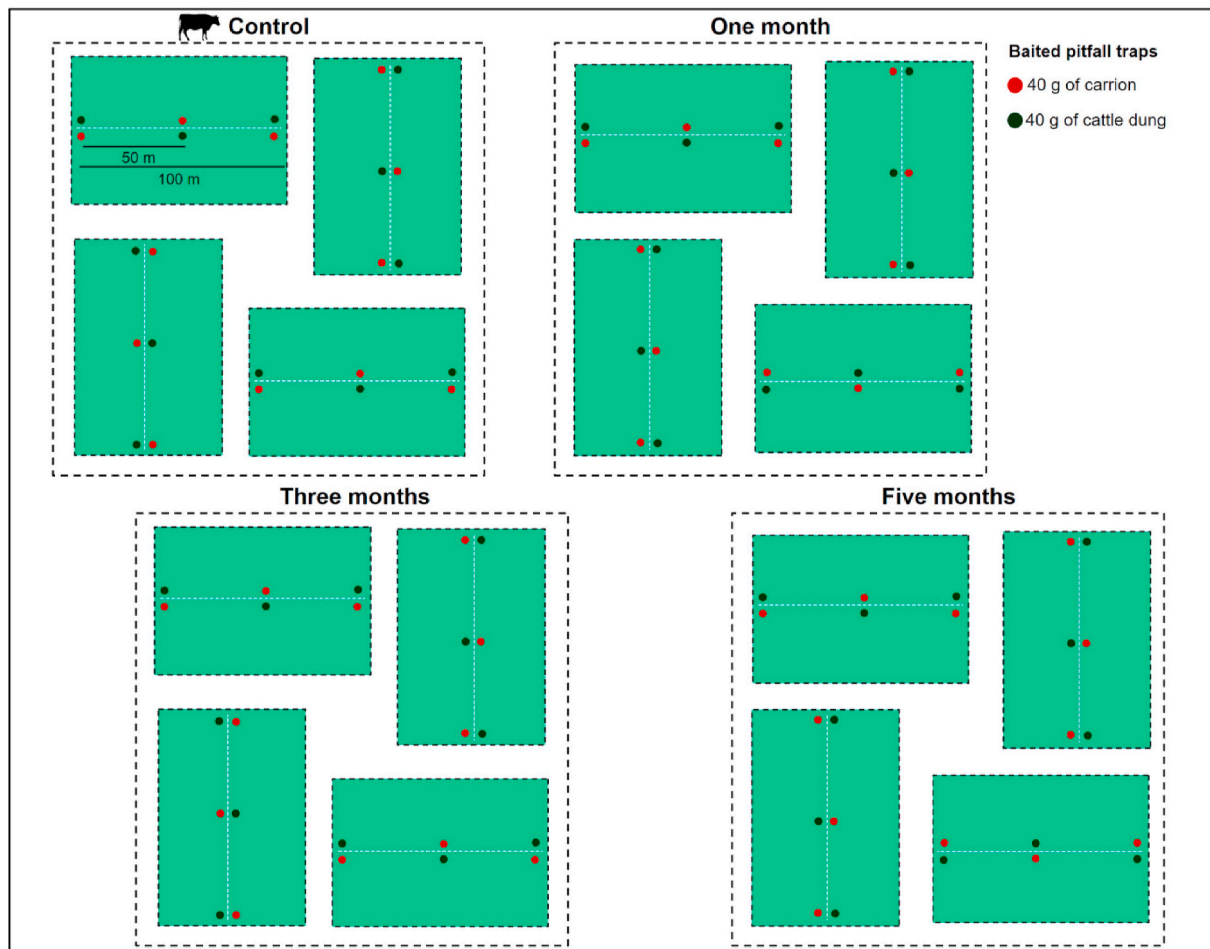


Fig. 1. Scheme of the sampling protocol of dung beetles employed in introduced Brazilian pastures with different times of cattle grazing removal, in Aquidauana, Mato Grosso do Sul, Brazil.

are rolled horizontally and then buried; tunneler species construct vertical tunnels below or adjacent to the food resource and transport dung into the bottom; and dweller species live within a dung pat and do not exhibit resource allocation. To analyze the diet, we used the proportion of individuals of each species attracted to a certain bait to determine bait specificity. A species was considered as “coprophage or necrophage” when its occurrence was $\geq 80\%$ on one bait (cattle dung or carrion), otherwise it was considered “generalist” (Halfpeter and Arellano, 2002). The minimum number of individuals of each species required to calculate diet was $n = 3$.

2.5. Data analysis

2.5.1. Species richness, number of individuals and biomass

The inventory completeness was evaluated using a sample coverage analysis (see Chao et al., 2014; Hsieh et al., 2016). This is a measure of sample completeness, and reveals the proportion of the total number of individuals in a community that belong to the species represented in the sample.

Data were first checked for normality using the Shapiro–Wilk test (Shapiro and Wilk, 1965) and for homoscedasticity using Bartlett’s test. We used Generalized Linear Models (GLMs) to test for differences in species richness, number of individuals and biomass (response variables) of dung beetles among pasture treatments with different cattle removal times (explanatory variables). We used Poisson errors corrected for overdispersion (quasi-Poisson) for dung beetle species richness; negative binomial errors for number of individuals; and Gaussian errors for biomass. All GLMs were submitted to residual analysis to evaluate

error distribution adequacy (Crawley, 2013). Analyses were conducted in R (R Development Core Team, 2020) with the package *lme4* for GLM.

2.5.2. Community composition

To verify differences in species composition among pasture treatments, we used Permutational Multivariate Anova (PERMANOVA) (Anderson, 2001). To test heterogeneity of multivariate dispersions between different pasture treatments, we used permutational multivariate analysis of dispersion (PERMDISP) (Anderson, 2001). We used a Non-Metric Multidimensional Scaling (NMDS) based on Bray-Curtis dissimilarity matrix with standardized and square-root transformed abundance species data to graphically represent the changes in dung beetle community composition among pasture treatments with different cattle removal times (Anderson and Willis, 2003). PERMANOVA, PERMDISP and NMDS analyses were implemented in the Primer v.6 software with PERMANOVA+ (Clarke and Gorley, 2006).

2.5.3. Functional diversity

We calculated community-weighted means of trait values (CWMs) and three functional diversity indexes that measure different aspects of functional diversity, using the R package *FD* in R software (R Development Core Team, 2020): 1) Functional dispersion (FDis) is estimated as the mean distance of all species to the abundance-weighted centroid of the community in the trait space (Laliberté and Legendre, 2010); 2) Functional evenness (FEve) summarizes how species abundances are distributed along the occupied functional space; and 3) Functional richness (FRic) represents the range of traits in a community quantified by the volume of functional trait space occupied (Villéger et al., 2008).

To evaluate differences in CWMs and three functional diversity indexes among pasture treatments, we used GLMs with Gaussian errors. All GLM were submitted to residual analysis to evaluate error distribution adequacy (Crawley, 2013), and conducted with the R package *lme4* in R software (R Development Core Team, 2020).

3. Results

3.1. Dung beetle species richness and number of individuals

We collected 563 dung beetle individuals belonging to 24 species of 13 genera and four tribes (Table 1). In the control system we recorded 11 species (134 individuals), while in cattle removal treatments we recorded 16 species and 254 individuals (one month); 18 species and 102 individuals (three months) and 14 species and 73 individuals (five months) (Table 1). We recorded high sample coverage estimates, ranging from 93% in three months to 99% in pastures with one month of cattle removal (Table 1). This indicates that the majority of the individuals in a community belong to species that were represented in our samples. We can thus consider our samples to be complete, and we have utilized empirical data for the diversity analysis and comparisons.

Ontherus appendiculatus (Mannerheim, 1829) and *Trichillum externepunctatum* Preudhomme de Borre, 1880 were the most abundant species in our study, representing 39 and 18% respectively, of all individuals sampled (Fig. 2). Species richness ($F_{3,12} = 0.62$, $p = 0.61$; Fig. 3A), number of individuals ($\chi^2_{3,12} = 2.14$, $p = 0.54$; Fig. 3B), biomass ($F_{3,12} = 2.17$, $p = 0.14$; Fig. 3C), Shannon ($F_{3,12} = 0.32$, $p = 0.80$; Fig. 3D) and Simpson diversity index ($F_{3,12} = 0.08$, $p = 0.96$; Fig. 3E) did not statistically differ among pasture treatments and control.

3.2. Community composition and dispersion patterns

Community composition did not differ among pasture treatments and control (Pseudo- $F = 1.33$, $p = 0.15$) (Fig. S1). In addition, pasture

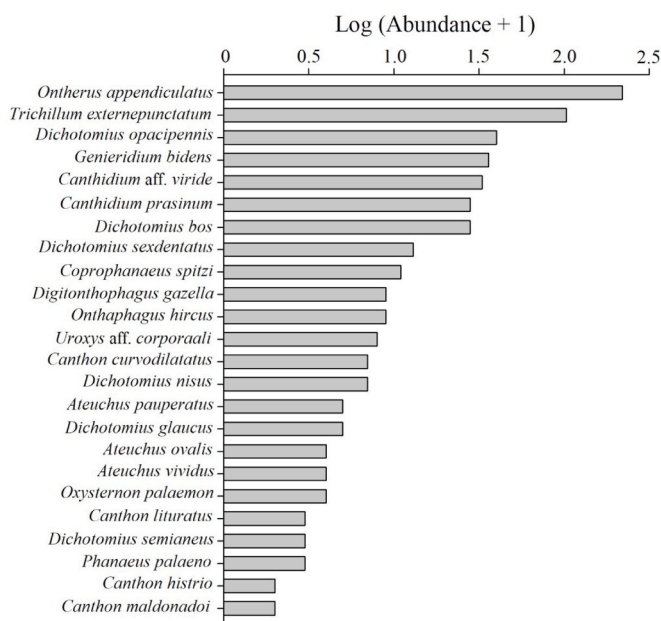


Fig. 2. Rank-abundance distribution in log (X+1) of dung beetle species sampled in introduced Brazilian pastures, in Aquidauana, Mato Grosso do Sul, Brazil.

treatments showed no differences in the multivariate dispersion of points (PERMDISP, $F = 2.42$, $p = 0.36$).

3.3. Functional diversity

Functional richness ($F_{3,12} = 0.34$, $p = 0.79$), functional dispersion ($F_{3,12} = 0.43$, $p = 0.73$) and functional evenness ($F_{3,12} = 0.64$, $p = 0.60$)

Table 1

Dung beetle species sampled in introduced Brazilian pastures with different times of cattle grazing removal in Aquidauana, Mato Grosso do Sul, Brazil. FG (Functional Group): Food relocation behavior (Rollers – R, Tunnellers – T, Dwellers – D, and missing data - NA).

Tribe	Species	FG	Control	Cattle removal (months)			Total
				One	Three	Five	
Ateuchini	<i>Ateuchus ovalis</i> Boucomont, 1928	T	0	1	0	2	3
	<i>Ateuchus pauperatus</i> (Galdino et al., 2016)	T	0	4	0	0	4
	<i>Ateuchus vividus</i> Galdino et al., 2016	T	0	2	1	0	3
	<i>Generidium bidens</i> (Balthasar, 1938)	D	14	10	1	10	35
	<i>Trichillum externepunctatum</i> Preudhomme de Borre, 1880	D	32	42	14	14	102
	<i>Uroxys aff. corporaali</i> Balthasar, 1940	NA	0	0	6	1	7
Copriini	<i>Ontherus appendiculatus</i> (Mannerheim, 1829)	T	64	139	9	7	219
Deltotrichilini	<i>Canthon curvodilatatus</i> Schmidt, 1920	R	2	1	2	1	6
	<i>Canthon histrio</i> Lepeletier and Serville, 1828	R	0	0	0	1	1
	<i>Canthon lituratus</i> (Germar, 1813)	R	0	0	2	0	2
	<i>Canthon maldonadoi</i> Martínez, 1951	R	0	0	1	0	1
Dichotomiini	<i>Canthidium aff. viride</i> (Lucas, 1857)	T	3	13	7	9	32
	<i>Canthidium prasinum</i> (Blanchard, 1846)	T	0	2	19	6	27
	<i>Dichotomius bos</i> (Blanchard, 1846)	T	5	15	2	5	27
	<i>Dichotomius glaucus</i> (Harold, 1869)	T	0	0	1	3	4
	<i>Dichotomius nissus</i> (Olivier, 1789)	T	2	0	2	2	6
	<i>Dichotomius opacipennis</i> (Luederwaldt, 1931)	T	0	0	29	10	39
	<i>Dichotomius semianus</i> (Galdino et al., 2016)	T	0	2	0	0	2
	<i>Dichotomius sexdentatus</i> (Luederwaldt, 1925)	T	1	10	1	0	12
	<i>Digitonthophagus gazella</i> (Fabricius, 1787)	T	7	0	1	0	8
Onthophagini	<i>Onthophagus hircus</i> Billberg, 1815	T	3	2	3	0	8
	<i>Coprophanaeus spitzii</i> (Pessôa, 1934)	T	0	7	1	2	10
Phanaeini	<i>Oxysternon palaemon</i> Castelnau, 1840	T	1	2	0	0	3
	<i>Phanaeus palaeno</i> Blanchard, 1846	T	0	2	0	0	2
Number of individuals (Abundance)							
Number of species (Richness)							
Sampling coverage (%)							
		134	254	102	73	563	
		11	16	18	14	24	
		99	99	93	96		

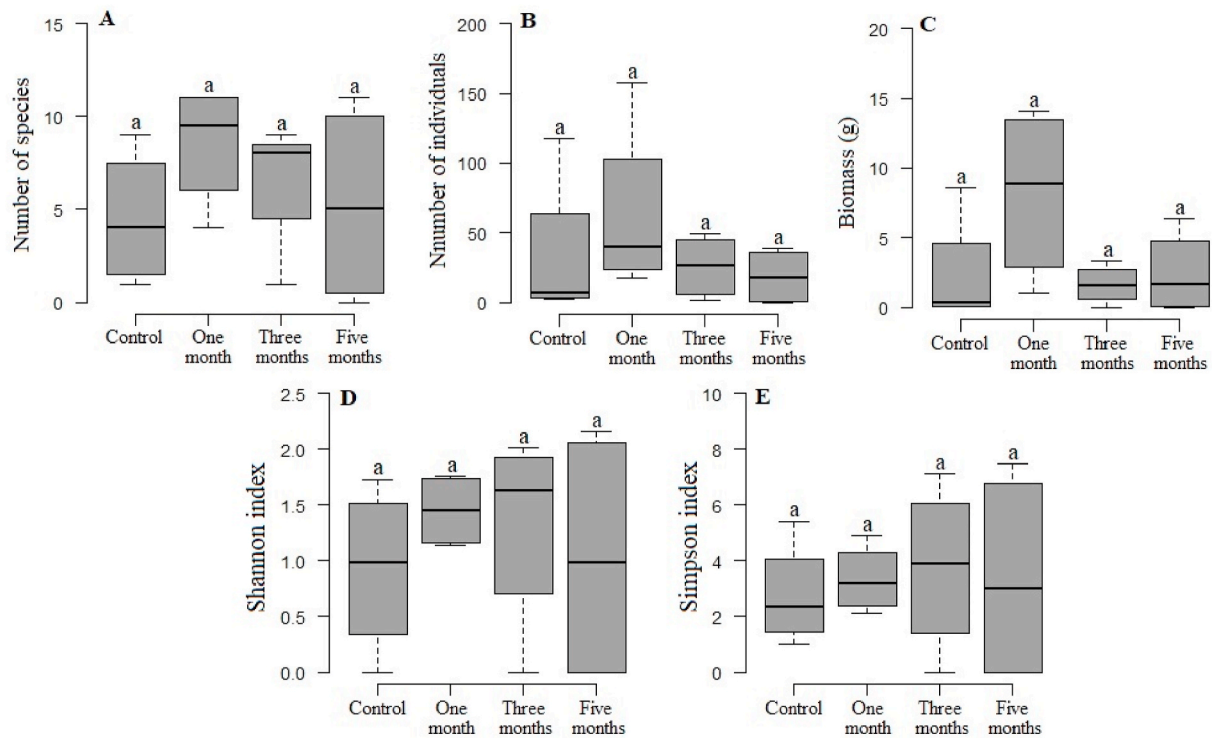


Fig. 3. Average species richness (A), average abundance (B), biomass (B), Shannon (D) and Simpson (E) indices of dung beetle sampled in introduced pastures with different times of cattle grazing removal in Aquidauana, Mato Grosso do Sul, Brazil. Identical letters above the bars indicate no statistical differences among pasture treatments and control ($p > 0.05$) according to F-test.

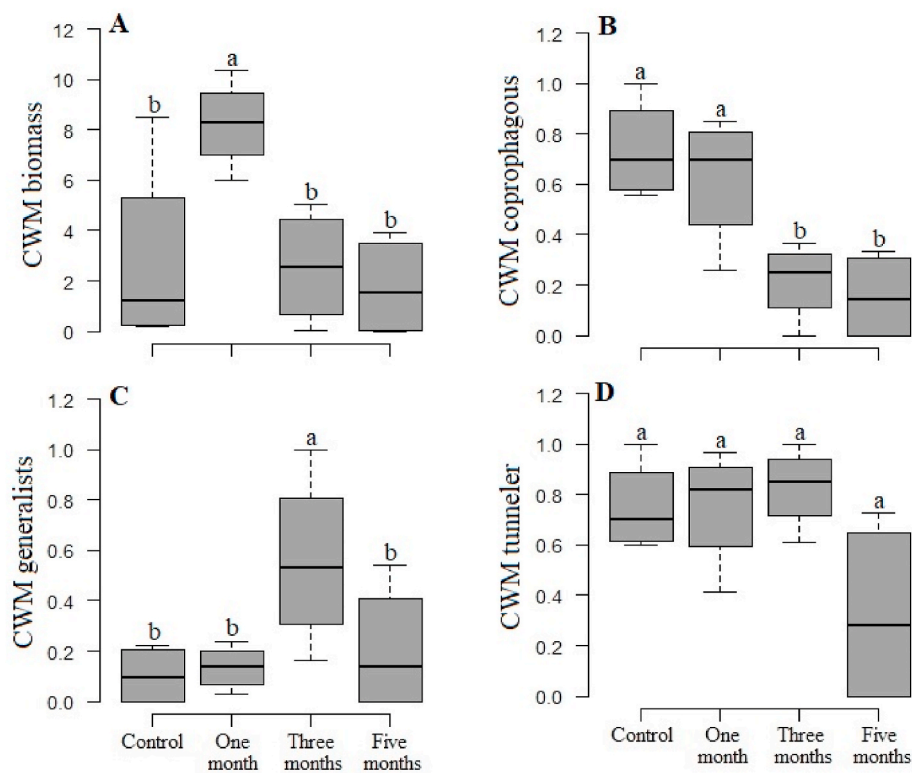


Fig. 4. Average CWM biomass (A), average CWM coprophagous (B), average CWM generalists (C) and CWM tunneler (D) of dung beetle assemblages sampled in introduced pastures with different times of cattle grazing removal in Aquidauana, Mato Grosso do Sul, Brazil. Different letters above the bars indicate statistically differences among pasture treatments and control ($p < 0.05$) according to F-test.

Table 2

Results of generalized linear models to test the effects of cattle grazing removal over a short-term on community-weighted means of trait values (CWMs) of dung beetle assemblages in introduced Brazilian pastures. Bold values are statistically significant ($p < 0.05$) according to F-test.

	F-test	p-value
CWM dweller	0.33	0.80
CWM roller	0.71	0.55
CWM tunneler	3.23	0.06
CWM coprophagous	8.18	0.003
CWM generalist	3.34	0.05
CWM necrophagous	0.41	0.74
CWM biomass	5.05	0.01

did not differ among pasture treatments and control (Fig. S2). However, CWM biomass was higher in pastures with one month of cattle removal in relation to control and other pasture treatments (Fig. 4A; Table 2). In addition, we found that CWM coprophagous diet are similar between control and pastures with one month of cattle removal, but decreased in pastures with three and five months of cattle removal (Fig. 4B; Table 2). CWM generalist diet was higher in pastures with three months of cattle removal (Fig. 4C), and we found a strong trend to decrease CWM tunneler nesting behavior in pastures with five months of cattle removal (Fig. 4D; Table 2). Fig. 3

4. Discussion

In this study, we evaluated, for the first time, the impacts of cattle grazing removal over a short-term on dung beetle assemblages in introduced Brazilian pastures. We demonstrated that despite cattle grazing removal, at least over the short-term, does not cause a negative impact on dung beetle taxonomic and functional diversity indexes, two important traits associated to dung removal are negatively affected after three months of cattle removal. Thus, we suggest the use of one month of cattle removal to grazing rotational management in introduced Brazilian pastures, besides conserving the grass, may maintain the similar taxonomic and functional diversity of dung beetles from the control, also conserving dung beetle traits that can aid in the dung removal, which indirectly influences the reduction of flies and parasitic worms (Braga et al., 2012; Nichols and Gómez, 2014) and soil fertilization rates (Nichols et al., 2008) in introduced pastures. Fig. 4

4.1. Patterns of species richness, number of individuals and biomass

In this study, 24 dung beetle species were found in introduced Brazilian pastures, which the number of surveyed species is congruent with what has been found before in the same region (i.e., Aida et al., 2000, $S = 19$; Correa et al., 2016b, $S = 19$; Correa et al., 2018, $S = 18$). Thus, our study contributes to understanding the diversity patterns and distribution of dung beetle species in introduced Brazilian pastures.

Ontherus appendiculatus and *Trichillum externepunctatum* were the most abundant species in the pastures studied. These species are widely distributed in introduced Brazilian pastures and considered of high importance for this ecosystem (Tissiani et al., 2017). Introduced pastures present adverse features for dung beetle colonization such as high temperatures, intense sun exposure, and compacted soils due to cattle trampling (Halffter et al., 1992; Navarrete and Halffter, 2008). However, dung beetle species are probably favored by forage capacity for cattle dung in exotic pastures, resulting in higher populations due to a competitive advantage for some habitat generalist species in this ecosystem (Louzada and Carvalho e Silva, 2009).

Dung beetle species richness, abundance, biomass and diversity (Shannon and Simpson indexes) did not differ among control pastures and different times after cattle grazing removal. Although dung beetles are dependent on mammals for their food resource (Raine and Slade, 2018), our results demonstrate that the presence of animal manure is not

the main factor to ensure a large number of dung beetle species and individuals in introduced pastures (Macedo et al., 2020). Thus, the greater availability of cattle dung is important, but not mandatory for the increase in richness and abundance of the local dung beetle assembly in open habitats (i.e., natural and introduced pastures) (Correa et al., 2020). Therefore, the absence of difference among pastures may be associated to cattle management performed by farmers in the region, which are very similar among proprietaries and can maintain some characteristics that affect dung beetles, such as: the grazing years, soil hardness, vegetation height (Imura et al., 2014), cattle density and frequency of application of veterinary medical products (Carvalho et al., 2020). Since dung beetles are bioindicators of environmental changes (Nichols et al., 2007), our results evidenced that cattle grazing removal *per se*, at least over the short-term ($\leq$ five months), did not cause a negative impact when compared with control pastures on dung beetle taxonomic metrics in introduced pastures.

4.2. Patterns of species composition

The dung beetle species composition and dispersion patterns did not differ among pasture treatments and control. Although introduced pastures are considered simple and homogeneous habitats (Louzada and Carvalho e Silva, 2009), variations in vegetation structure is an important filter to dung beetle assemblages in exotic pastures (Macedo et al., 2020), because directly modify the local microclimate conditions including variations in temperature and atmospheric humidity, surface soil temperature, as well as the degree of direct sunlight to potentially affect the dung beetle species composition (Larsen, 2012). Thus, this result implies that the environmental conditions and vegetation structure, after five months of cattle removal (short-term), were not different enough to modify the dung beetle species composition from the control pasture.

4.3. Functional diversity

We did not find differences in FRic, FDis, and FEve metrics among pasture treatments and control, following the same pattern recorded in taxonomic metrics. Overall, functional responses have been shown to depend mainly on the intensity of the disturbance and the functional characteristics chosen (Mlambo, 2014; Beiroz et al., 2018). Since a low intensity disturbance in highly diversified communities does not modify functional structure (Magnago et al., 2014), the removal of cattle grazing over short-term may represent a low disturbance for dung beetle functional diversity in introduced Brazilian pastures. In addition, the maintenance of dung beetle species richness, abundance, species composition and diversity from the control pastures may help to avoid the loss of functionally specialized species (Larsen et al., 2005; Barrágan et al., 2011; Audino et al., 2014), resulting in a lack of reduction in functional diversity after cattle removal.

Although cattle grazing removal, at least over the short-term, does not cause a negative effect on functional diversity indexes, CWM biomass and CWM coprophagous are negatively affected after three months of cattle removal. In addition, we observed a marginally significant trend to reduce CWM tunneler nesting behavior in pastures with five months of cattle removal. Indeed, CWM coprophagous was similar between control and pastures with one month of cattle removal, and CWM biomass found in pastures with one month of cattle removal was higher than that found in control pastures. Since that CWM biomass, coprophagous and tunneler nesting behavior are important traits associated to dung removal (Slade et al., 2007; Nervo et al., 2014), our results suggest that one month of cattle removal is more adequate to maintain dung beetle traits from the control pastures. Thus, we may conserve dung beetle traits that can aid in the dung removal, which indirectly influences the reduction of flies and parasitic worms (Braga et al., 2012; Nichols and Gómez, 2014) and soil fertilization rates (Nichols et al., 2008) in introduced pastures.

4.4. Implications of cattle grazing management on dung beetle conservation in introduced Brazilian pastures

Agroecosystems represent a large area in most terrestrial landscapes, and management decisions within these systems can affect the biodiversity conservation. In intensive grazing systems where cattle remain in the same pasture during the entire year, the continuous movement of these animals reduces vegetation cover since they may trample young plants or damage adult plants, as well as compress and destabilize the soil surface and increase areas of bare soil (Spavorek et al., 2007). These effects may increase soil erosion, and consequently the degradation of pastures that have potential to negatively affect the dung beetle assemblages. Thus, pasture management continuously produces forage with low, dense structure (Briske et al., 2008), with limited opportunities for the ecological niche of dung arthropod assemblages (Pecenka and Lundgren, 2019).

Recently, Pecenka and Lundgren (2019) found a higher dung beetle abundance in pastures with frequent rotational grazing when compared to pastures with continuous grazing. This occurs because rotational grazing can positively affect dung beetles in several ways. Rotational grazing alters resources available to cattle within a pasture, leading to less bare ground within the pasture over a four-year period (Jacobo et al., 2006). Thus, plants from pastures that are grazed intensively and then allowed to rest have high calcium, magnesium, and crude protein content (Walton et al., 1981; Teague et al., 2013), affecting directly the dung pat composition, and can lead to a more attractive resource for dung beetles (Holter and Scholtz, 2007). Finally, rotational grazing can be used for control of gastrointestinal nematodes (Barger et al., 1994; Rapiya et al., 2019), reducing the frequency of use the anthelmintic products that are detrimental for dung beetles (Verdú et al., 2015; Carvalho et al., 2020).

4.5. Caveats

We are aware that there is evidence of some species of dung beetles that could move more than 1 km per day (Arellano et al., 2008). Although we tried to maximize the distance among pasture treatments (with a minimal distance of 300 m) it was so difficult to guarantee a total spatial independence between sites (i.e., a distance of ± 1 km) due to the structure organization of the sites. However, in this livestock system the usual distance between rotational pastures is usually less than 300 m. Moreover, it is important to mention that the use of cattle dung in a livestock system could affect our results in terms of potential competition in a landscape perspective, taking into consideration the high spatial supply of this kind of resource. However, we believe that our traps were not affected by this possible effect because in the control pastures we did not register any differences with the treatments in terms of dung beetle abundance and richness. Furthermore, the use of carrion, might homogenize the abundance, richness, and composition of dung beetles in the pasture treatments. It is possible that the species that have a strong tendency to be necrophagous (e.g., *Ateuchus ovalis* Boucomont, 1928, *Canthon curvodilatatus* Schmidt, 1920, *Canthidium prasinum* (Blanchard, 1846), *Coprophanaeus spitzi* (Pessôa, 1934), *Dichotomius sexdentatus* (Luederwaldt, 1925), and *Oxysternon palaemon* Castelnau, 1840) could not change even in response to cattle removal from the pastures, because we were offering a different resource in the pitfall traps. In this sense, because we found differences in CWM value of traits, it is important to recommend to perform further studies to evaluate the functional effect of this cattle removal in terms of the ecosystem services. Also, we are aware of the short temporal scale of evaluation that we selected (i.e., five months). However, on this region this is the maximum average value of cattle removal or rotation time between pastures. Therefore, it seemed an appropriate time, although it would be interesting in future studies to explore longer removal times (e.g., 9, 12, 24 months), taking in consideration that some studies (Escobar et al., 2008) propose that changes in the structure of dung beetle assemblages

under human modified landscapes can be observed over a long term.

5. Conclusions

Our results provide evidence based on taxonomic and functional diversity indexes that cattle grazing removal *per se*, at least over a short-term (i.e., five months), does not cause a strong negative impact on dung beetle biodiversity in introduced Brazilian pastures. However, two important traits associated to dung removal are negatively affected after three months of cattle removal. As such, we show that in introduced Brazilian pastures a rotational grazing up to one month can maintain the same taxonomic and functional metrics of dung beetles acting on these pastures when both soil and grasses are recovering from cattle grazing and trampling. Thus, we suggest that one month of cattle removal to grazing rotational management in introduced Brazilian pastures may be a useful strategy to, besides conserving the grass of the pastures and the surrounding forest cover (Nichols et al., 2007; Numa et al., 2009; Eaton et al., 2011), also conserving dung beetle biodiversity (Navarrete and Halfpeter, 2008; Audino et al., 2014; Pecenka and Lundgren, 2019) and consequently the ecological functions performed by them.

CRediT authorship contribution statement

César M.A. Correa: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing - original draft, Writing - review & editing. **Marco A. Lara:** Methodology, Investigation, Writing - original draft. **Anderson Puker:** Conceptualization, Methodology, Writing - original draft, Writing - review & editing. **Jorge Ari Noriega:** Validation, Writing - original draft, Writing - review & editing. **Vanessa Korasaki:** Supervision, Conceptualization, Methodology, Validation, Writing - original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.actao.2020.103681>.

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